

Rust gRPC servicers vs Python servicers vs native HTTP: SGLang, vLLM and TokenSpeed on nvidia/Qwen3.8-27B-NVFP4

Model. nvidia/Qwen3.8-27B-NVFP4 (a Qwen3.5-family hybrid-attention model, mixed FP8/NVFP4 checkpoint), one GPU per replica, every engine with its own default settings.

Host. One aarch64 host with 4 GPUs, all software and the model on local disk. Only one worker set is alive at a time; each stack ran one lifetime with 2 passes, so every table cell is the mean of up to 2 runs (the spread column is (max-min)/mean of requests per second across those runs).

Engines. SGLang 0.5.20 (the CI pin), vLLM nightly 0.30.1rc1.dev640 (torch 2.13, CUDA 13), TokenSpeed nightly 0.1.0.post20261004 (torch 2.14, CUDA 13, FlashInfer 0.7.1rc2). Engine defaults everywhere: the command lines carry the model, host, port and the endpoint selector, nothing else. One exception, needed to run at all: the 4-replica TokenSpeed workers get --kvstore-size 120 (GB of host cache each) because the default sizes the host cache from free RAM and four of them oversubscribe the host.

smg. The gateway runs in gRPC mode in front of the engines' gRPC servicers and in HTTP mode in front of the native servers, with the tokenizer cache on (--tokenizer-cache-enable-l0 --tokenizer-cache-enable-l1) and the default routing policy. Python servicers: `python -m sglang.launch_server --grpc-mode, vllm serve --grpc, python -m smg_grpc_servicer.tokenspeed`. Rust servicers: the same endpoints with `SMG_<ENGINE>_SERVICER_IMPL=rust` (smg's Rust servicer takes the gRPC port and drives the engine over its own in-process wire).

Baselines. The SGLang and vLLM native HTTP servers (`python -m sglang.launch_server, vllm serve`) and vLLM's Rust frontend (`VLLM_USE_RUST_FRONTEND=1 vllm serve`), each measured directly and behind smg HTTP mode at 1 replica, and behind smg HTTP mode at 4 replicas (where a load balancer is needed).

Load. /v1/completions, streaming, ignore_eos so every request produces exactly the requested output length, each request carrying its own prompt pinned to the input length (no prefix-cache reuse across requests), temperature 0.7, token counts from the stream's usage. A shape is clients x input tokens x output tokens; a warm-up run precedes every measurement. TTFT is time to the first streamed token, TPOT the mean time per output token after the first, end-to-end the whole request. CPU seconds are the host CPU time consumed over the run by the worker's whole process tree (engine plus servicer) and by the smg process. A request that fails is counted and excluded from the latency statistics; cells with failures are marked.

SGLang, 1 replica, Rust servicer vs Python servicer (geometric mean of the per-shape ratios over 9 shapes): throughput x1.000, TPOT p50 x0.997, TTFT p99 x0.97, worker-tree CPU x0.56, smg CPU x1.07.

vLLM, 1 replica, Rust servicer vs Python servicer (geometric mean of the per-shape ratios over 9 shapes): throughput x1.002, TPOT p50 x0.997, TTFT p99 x0.98, worker-tree CPU x0.63, smg CPU x1.17.

TokenSpeed, 1 replica, Rust servicer vs Python servicer (geometric mean of the per-shape ratios over 9 shapes): throughput x1.002, TPOT p50 x1.000, TTFT p99 x0.96, worker-tree CPU x1.05, smg CPU x1.03.

SGLang, 1 replica, relative to its native HTTP server, direct: behind smg HTTP mode: throughput x0.999, TTFT p99 x1.01, TPOT p50 x1.001 (9 shapes); smg gRPC mode with the Python servicer: throughput x0.996, TTFT p99 x1.02, TPOT p50 x1.015 (9 shapes); smg gRPC mode with the Rust servicer: throughput x0.996, TTFT p99 x0.98, TPOT p50 x1.012 (9 shapes).

vLLM, 1 replica, relative to its native HTTP server, direct: behind smg HTTP mode: throughput x1.001, TTFT p99 x0.98, TPOT p50 x0.996 (9 shapes); smg gRPC mode with the Python servicer: throughput x0.984, TTFT p99 x1.01, TPOT p50 x1.013 (9 shapes); smg gRPC mode with the Rust servicer: throughput x0.986, TTFT p99 x0.99, TPOT p50 x1.009 (9 shapes); vLLM Rust frontend, direct: throughput x0.997, TTFT p99 x1.04, TPOT p50 x1.027 (9 shapes); vLLM Rust frontend behind smg HTTP mode: throughput x1.001, TTFT p99 x0.98, TPOT p50 x1.024 (9 shapes).

Requests / s, 1 replica

shape (clients-input-output)	SGLang Python service	SGLang Rust service	vLLM Python service	vLLM Rust service	TokenSpeed Python service	TokenSpeed Rust service	SGLang HTTP direct	SGLang HTTP + smg	vLLM HTTP direct	vLLM HTTP + smg	LM Rust frontend direct	LM Rust frontend + smg
c1-i512-o128	1.25	1.28	1.21	1.21	1.15	1.15	1.28	1.28	1.21	1.21	1.21	1.22
c8-i256-o128	8.75	8.77	7.47	7.30	7.89	8.03	8.74	8.75	7.49	7.38	7.30	7.33
c16-i512-o2048	1.06	1.06	1.04	1.04	0.96	0.96	1.07	1.06	1.04	1.04	1.04	1.04
c32-i1024-o256	10.88	10.92	10.86	10.82	10.42	10.61	10.86	10.84	10.77	10.80	10.65	10.37
c32-i4096-o256	5.56	5.58	6.38	6.45	6.08	6.04	5.59	5.59	6.42	6.45	6.46	6.55
c64-i2048-o1024	4.09	4.09	4.46	4.47	4.27	4.25	4.11	4.11	4.44	4.47	4.39	4.46
c64-i8192-o256	3.27	3.29	4.71	4.72	4.21	4.22	3.29	3.29	4.71	4.72	4.68	4.72
c128-i1024-o256	17.62	17.16	19.78	19.89	17.05	16.84	17.59	17.42	19.52	19.80	20.36	20.23
c256-i1024-o256	18.81	18.73	19.69	20.04	18.85	18.95	18.78*	18.82	23.09	22.91	22.72	23.02

Output tokens / s, 1 replica

shape (clients-input-output)	SGLang Python service	SGLang Rust service	vLLM Python service	vLLM Rust service	TokenSpeed Python service	TokenSpeed Rust service	SGLang HTTP direct	SGLang HTTP + smg	vLLM HTTP direct	vLLM HTTP + smg	LM Rust frontend direct	LM Rust frontend + smg
c1-i512-o128	160	164	154	155	148	148	164	164	155	154	155	156
c8-i256-o128	1,119	1,122	957	935	1,010	1,028	1,119	1,120	958	946	935	938
c16-i512-o2048	2,174	2,178	2,134	2,132	1,970	1,967	2,185	2,183	2,134	2,132	2,123	2,132
c32-i1024-o256	2,785	2,795	2,780	2,771	2,668	2,714	2,779	2,775	2,756	2,767	2,727	2,653
c32-i4096-o256	1,425	1,428	1,634	1,651	1,556	1,548	1,432	1,431	1,642	1,650	1,654	1,677
c64-i2048-o1024	4,193	4,184	4,568	4,581	4,372	4,356	4,208	4,203	4,548	4,581	4,502	4,561
c64-i8192-o256	839	841	1,205	1,208	1,080	1,081	844	843	1,207	1,210	1,199	1,210
c128-i1024-o256	4,510	4,392	5,063	5,091	4,365	4,311	4,502	4,461	4,997	5,068	5,214	5,178
c256-i1024-o256	4,817	4,796	5,041	5,128	4,825	4,852	4,808*	4,819	5,910	5,863	5,816	5,891

TTFT p50 (ms), 1 replica

shape (clients-input-output)	GLang Python service	SGLang Rust service	vLLM Python service	vLLM Rust service	TokenSpeed Python service	TokenSpeed Rust service	SGLang HTTP direct	SGLang HTTP + smg	vLLM HTTP direct	vLLM HTTP + smg	LM Rust frontend direct	LM Rust frontend + smg
c1-i512-o128	68	65	60	54	52	52	66	69	58	60	53	51
c8-i256-o128	127	122	228	246	113	92	128	129	228	242	239	237
c16-i512-o2048	241	236	224	231	210	217	242	245	236	242	300	233
c32-i1024-o256	532	526	752	788	400	382	747	746	777	794	793	876
c32-i4096-o256	1,763	1,737	1,740	1,719	1,436	1,440	1,744	1,741	1,607	1,544	1,050	1,019
c64-i2048-o1024	1,661	1,733	1,464	1,566	1,407	1,419	1,719	1,761	1,382	1,344	1,157	1,113
c64-i8192-o256	7,535	7,468	2,214	2,069	5,567	5,571	7,460	7,489	1,933	2,075	1,514	1,507
c128-i1024-o256	1,593	1,823	1,371	1,358	1,715	1,822	1,518	1,545	1,281	1,381	1,104	1,078
c256-i1024-o256	8,174	8,277	1,344	1,314	5,369	5,341	8,322*	8,332	1,236	1,398	1,214	1,442

TTFT p99 (ms), 1 replica

shape (clients-input-output)	GLang Python service	SGLang Rust service	vLLM Python service	vLLM Rust service	TokenSpeed Python service	TokenSpeed Rust service	SGLang HTTP direct	SGLang HTTP + smg	vLLM HTTP direct	vLLM HTTP + smg	LM Rust frontend direct	LM Rust frontend + smg
c1-i512-o128	111	106	92	87	89	93	105	107	101	100	94	86
c8-i256-o128	149	141	295	323	143	147	149	150	263	295	365	360
c16-i512-o2048	267	245	255	268	248	250	249	270	274	278	385	264
c32-i1024-o256	778	772	1,030	977	1,203	722	796	802	1,077	991	1,080	1,153
c32-i4096-o256	3,157	3,143	2,739	2,668	2,672	2,678	3,140	3,141	2,685	2,591	2,655	2,526
c64-i2048-o1024	2,992	2,976	2,648	2,564	2,666	2,677	2,980	2,972	2,827	2,620	2,770	2,566
c64-i8192-o256	14,379	14,312	10,124	10,082	10,904	10,888	14,282	14,312	10,043	9,991	10,180	9,982
c128-i1024-o256	3,419	3,322	2,764	2,766	3,157	3,279	3,437	3,380	3,184	2,789	2,702	2,595
c256-i1024-o256	10,573	10,168	7,727	6,916	10,180	10,158	10,322*	10,160	5,585	5,546	5,404	5,422

TPOT p50 (ms), 1 replica

shape (clients-input-output)	SGLang Python service	SGLang Rust service	vLLM Python service	vLLM Rust service	TokenSpeed Python service	TokenSpeed Rust service	SGLang HTTP direct	SGLang HTTP + smg	vLLM HTTP direct	vLLM HTTP + smg	LM Rust frontend direct	LM Rust frontend + smg
c1-i512-o128	5.77	5.63	6.05	6.06	6.41	6.41	5.62	5.62	6.05	6.05	6.05	6.05
c8-i256-o128	6.19	6.20	6.58	6.59	7.05	7.04	6.18	6.18	6.58	6.57	6.59	6.63
c16-i512-o2048	7.24	7.23	7.38	7.39	8.02	8.02	7.21	7.21	7.38	7.39	7.38	7.39
c32-i1024-o256	9.45	9.45	8.41	8.42	10.30	10.30	8.56	8.57	8.42	8.38	8.42	8.43
c32-i4096-o256	15.60	15.66	12.65	12.65	14.94	14.99	15.57	15.59	13.23	13.46	15.28	15.20
c64-i2048-o1024	13.63	13.52	12.59	12.46	13.21	13.26	13.48	13.50	12.67	12.62	13.02	12.94
c64-i8192-o256	46.94	47.23	43.66	44.19	37.48	37.50	46.86	46.89	45.02	44.28	47.19	47.05
c128-i1024-o256	21.71	21.54	19.71	19.84	22.30	22.20	22.30	22.46	20.28	19.75	20.24	20.41
c256-i1024-o256	21.98	22.11	47.20	45.36	28.87	28.84	21.73*	21.61	37.78	37.69	38.66	37.52

End-to-end p99 (ms), 1 replica

shape (clients-input-output)	GLang Python service	SGLang Rust service	vLLM Python service	vLLM Rust service	TokenSpeed Python service	TokenSpeed Rust service	SGLang HTTP direct	SGLang HTTP + smg	vLLM HTTP direct	vLLM HTTP + smg	LM Rust frontend direct	LM Rust frontend + smg
c1-i512-o128	844	822	860	856	903	910	814	817	868	868	894	855
c8-i256-o128	937	931	1,130	1,159	1,053	1,067	934	932	1,097	1,129	1,201	1,197
c16-i512-o2048	15,096	15,083	15,382	15,396	16,678	16,699	15,029	15,032	15,386	15,403	15,506	15,401
c32-i1024-o256	2,952	2,945	3,176	3,126	3,555	3,086	2,976	2,981	3,209	3,120	3,305	3,329
c32-i4096-o256	5,762	5,747	6,091	5,971	5,314	5,380	5,743	5,738	6,129	6,032	6,707	6,645
c64-i2048-o1024	15,694	15,792	15,804	15,535	15,035	15,134	15,641	15,658	16,087	15,800	16,326	16,082
c64-i8192-o256	19,571	19,552	21,732	21,756	15,255	15,253	19,486	19,497	21,943	21,704	22,679	22,449
c128-i1024-o256	8,170	7,809	7,931	7,997	7,956	8,078	7,864	7,852	8,668	7,827	8,206	8,044
c256-i1024-o256	15,682	15,379	20,279	19,027	17,267	17,140	15,238*	15,070	14,982	15,249	15,240	14,975

Worker-tree CPU (s), 1 replica

shape (clients-input-output)	SGLang Python service	SGLang Rust service	vLLM Python service	vLLM Rust service	TokenSpeed Python service	TokenSpeed Rust service	SGLang HTTP direct	SGLang HTTP + smg	vLLM HTTP direct	vLLM HTTP + smg	LM Rust frontend direct	LM Rust frontend + smg
c1-i512-o128	1.0	0.5	0.8	0.6	0.9	0.9	0.8	1.0	1.1	1.0	0.0	0.0
c8-i256-o128	1.2	0.7	1.0	0.8	1.2	1.3	1.2	1.2	1.3	1.3	0.0	0.0
c16-i512-o2048	7.3	4.6	6.4	5.0	7.8	8.1	6.1	6.5	6.1	6.0	0.0	0.0
c32-i1024-o256	5.3	2.8	4.8	3.7	5.8	5.8	5.4	5.8	5.9	6.4	0.0	0.0
c32-i4096-o256	4.2	2.4	3.6	2.6	3.8	4.6	6.2	6.1	7.7	7.2	0.0	0.0
c64-i2048-o1024	10.1	4.6	8.8	5.2	10.8	10.6	9.2	9.8	9.1	8.7	0.0	0.0
c64-i8192-o256	2.0	1.2	1.7	1.1	2.0	2.6	4.6	5.1	6.0	6.0	0.0	0.0
c128-i1024-o256	13.8	9.0	14.8	5.5	15.7	13.3	17.0	17.0	16.7	16.9	0.0	0.0
c256-i1024-o256	28.6	17.6	31.6	13.8	32.6	35.2	34.9*	35.3	30.0	30.2	0.0	0.0

smg CPU (s), 1 replica

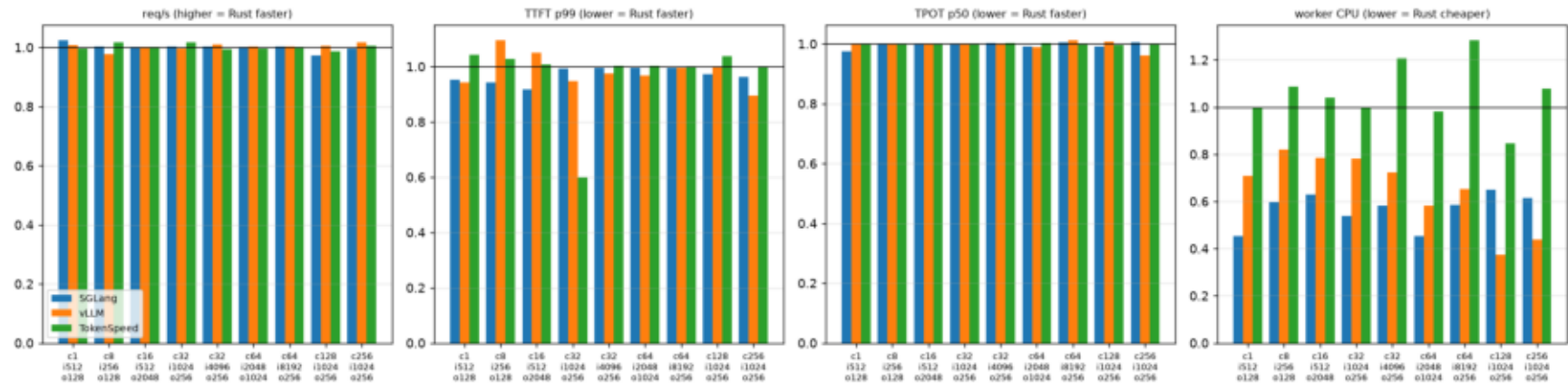
shape (clients-input-output)	SGLang Python service	SGLang Rust service	vLLM Python service	vLLM Rust service	TokenSpeed Python service	TokenSpeed Rust service	SGLang HTTP direct	SGLang HTTP + smg	vLLM HTTP direct	vLLM HTTP + smg	LM Rust frontend direct	LM Rust frontend + smg
c1-i512-o128	0.5	0.5	0.6	0.6	0.6	0.6	0.0	0.4	0.0	0.5	0.0	0.4
c8-i256-o128	1.3	1.4	1.4	1.3	1.3	1.5	0.0	1.0	0.0	1.0	0.0	1.0
c16-i512-o2048	6.9	6.9	5.6	7.0	6.4	6.9	0.0	6.1	0.0	6.9	0.0	5.4
c32-i1024-o256	7.1	6.9	6.4	7.2	6.5	7.2	0.0	4.4	0.0	4.3	0.0	4.1
c32-i4096-o256	9.3	9.2	9.1	9.5	9.2	9.1	0.0	3.2	0.0	4.0	0.0	2.7
c64-i2048-o1024	10.9	11.8	10.1	13.9	10.7	11.3	0.0	7.6	0.0	7.7	0.0	7.8
c64-i8192-o256	7.4	7.7	7.3	7.8	7.4	7.0	0.0	1.4	0.0	2.2	0.0	1.7
c128-i1024-o256	18.1	22.4	16.4	25.0	18.3	20.1	0.0	10.8	0.0	13.0	0.0	10.0
c256-i1024-o256	39.0	49.6	35.1	49.2	38.3	35.0	0.0*	24.6	0.0	26.2	0.0	22.5

Rust servicer / Python servicer (ratios), 1 replica

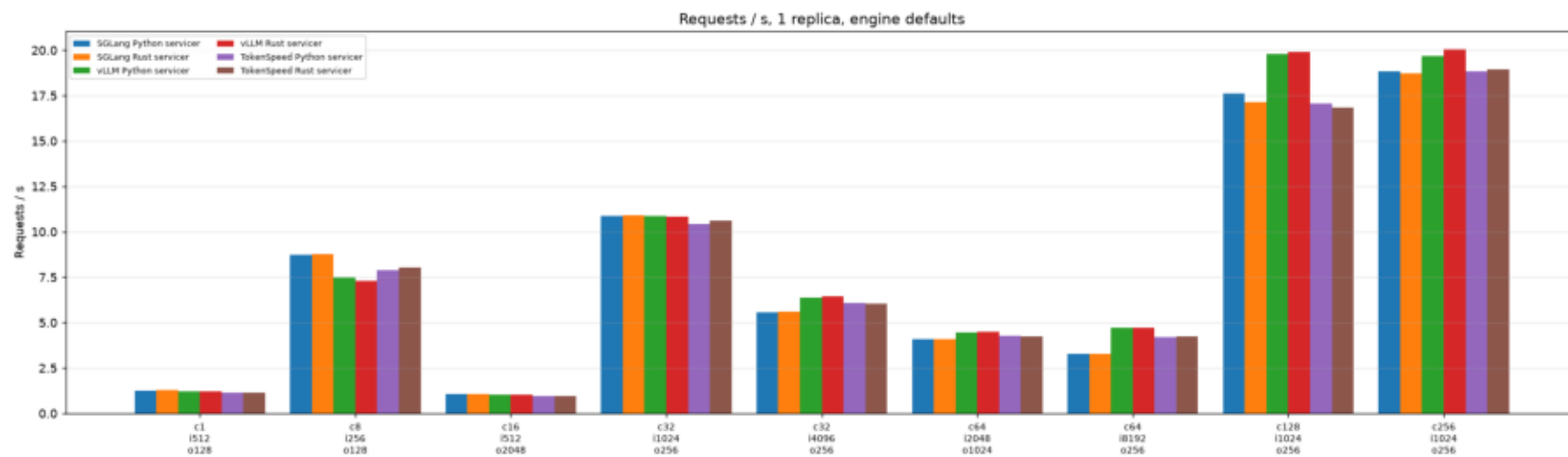
engine	shape	req/s	out tok/s	TTFT p50	TTFT p99	TPOT p50	e2e p99	worker CPU	smg CPU
SGLang	c1-i512-o128	1.02	1.03	0.96	0.95	0.98	0.97	0.45	1.03
vLLM	c1-i512-o128	1.01	1.01	0.89	0.94	1.00	1.00	0.71	0.94
TokenSpeed	c1-i512-o128	1.00	1.00	1.00	1.04	1.00	1.01	1.00	0.98
SGLang	c8-i256-o128	1.00	1.00	0.97	0.94	1.00	0.99	0.60	1.08
vLLM	c8-i256-o128	0.98	0.98	1.08	1.10	1.00	1.03	0.82	0.92
TokenSpeed	c8-i256-o128	1.02	1.02	0.81	1.03	1.00	1.01	1.09	1.17
SGLang	c16-i512-o2048	1.00	1.00	0.98	0.92	1.00	1.00	0.63	1.00
vLLM	c16-i512-o2048	1.00	1.00	1.03	1.05	1.00	1.00	0.78	1.26
TokenSpeed	c16-i512-o2048	1.00	1.00	1.03	1.01	1.00	1.00	1.04	1.07
SGLang	c32-i1024-o256	1.00	1.00	0.99	0.99	1.00	1.00	0.54	0.98
vLLM	c32-i1024-o256	1.00	1.00	1.05	0.95	1.00	0.98	0.78	1.12
TokenSpeed	c32-i1024-o256	1.02	1.02	0.96	0.60	1.00	0.87	1.00	1.10
SGLang	c32-i4096-o256	1.00	1.00	0.99	1.00	1.00	1.00	0.58	0.99
vLLM	c32-i4096-o256	1.01	1.01	0.99	0.97	1.00	0.98	0.72	1.05
TokenSpeed	c32-i4096-o256	0.99	1.00	1.00	1.00	1.00	1.01	1.21	0.99
SGLang	c64-i2048-o1024	1.00	1.00	1.04	0.99	0.99	1.01	0.45	1.08
vLLM	c64-i2048-o1024	1.00	1.00	1.07	0.97	0.99	0.98	0.58	1.38
TokenSpeed	c64-i2048-o1024	1.00	1.00	1.01	1.00	1.00	1.01	0.98	1.05
SGLang	c64-i8192-o256	1.00	1.00	0.99	1.00	1.01	1.00	0.58	1.04
vLLM	c64-i8192-o256	1.00	1.00	0.93	1.00	1.01	1.00	0.65	1.07
TokenSpeed	c64-i8192-o256	1.00	1.00	1.00	1.00	1.00	1.00	1.28	0.95
SGLang	c128-i1024-o256	0.97	0.97	1.14	0.97	0.99	0.96	0.65	1.24
vLLM	c128-i1024-o256	1.01	1.01	0.99	1.00	1.01	1.01	0.37	1.52
TokenSpeed	c128-i1024-o256	0.99	0.99	1.06	1.04	1.00	1.02	0.85	1.10
SGLang	c256-i1024-o256	1.00	1.00	1.01	0.96	1.01	0.98	0.61	1.27
vLLM	c256-i1024-o256	1.02	1.02	0.98	0.90	0.96	0.94	0.44	1.40
TokenSpeed	c256-i1024-o256	1.01	1.01	0.99	1.00	1.00	0.99	1.08	0.91

Rust servicer relative to the Python servicer, 1 replica

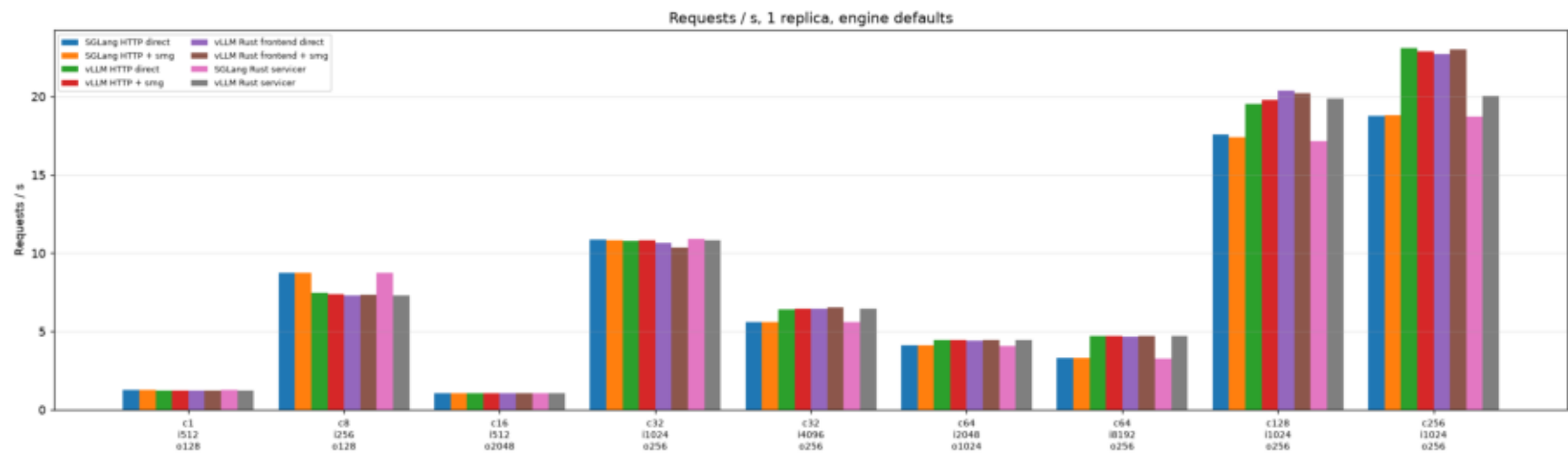
Rust servicer / Python servicer, same engine and gateway, 1 replica (1.0 = equal)



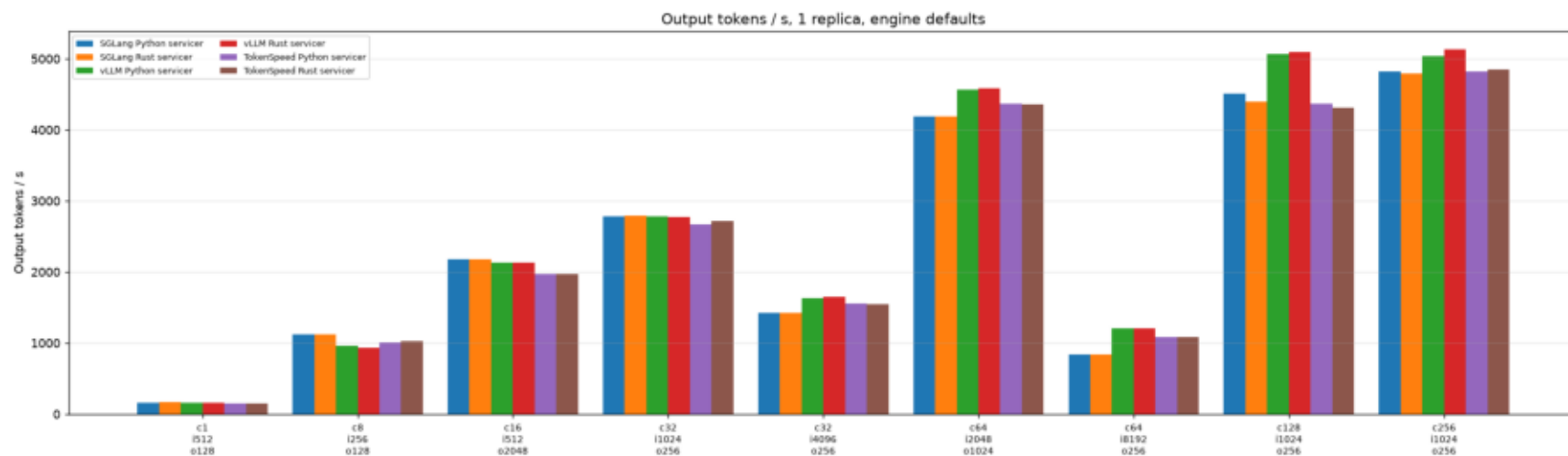
Requests / s, gRPC servicers behind smg gRPC mode, 1 replica



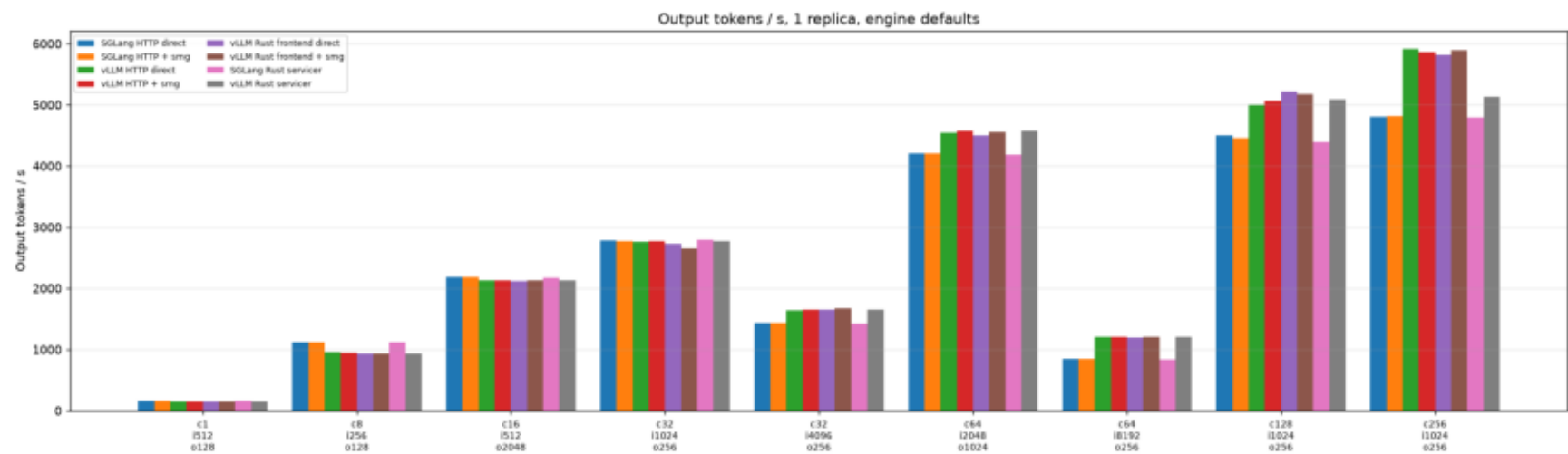
Requests / s, native servers direct vs behind smg vs smg gRPC with the Rust servicer, 1 replica



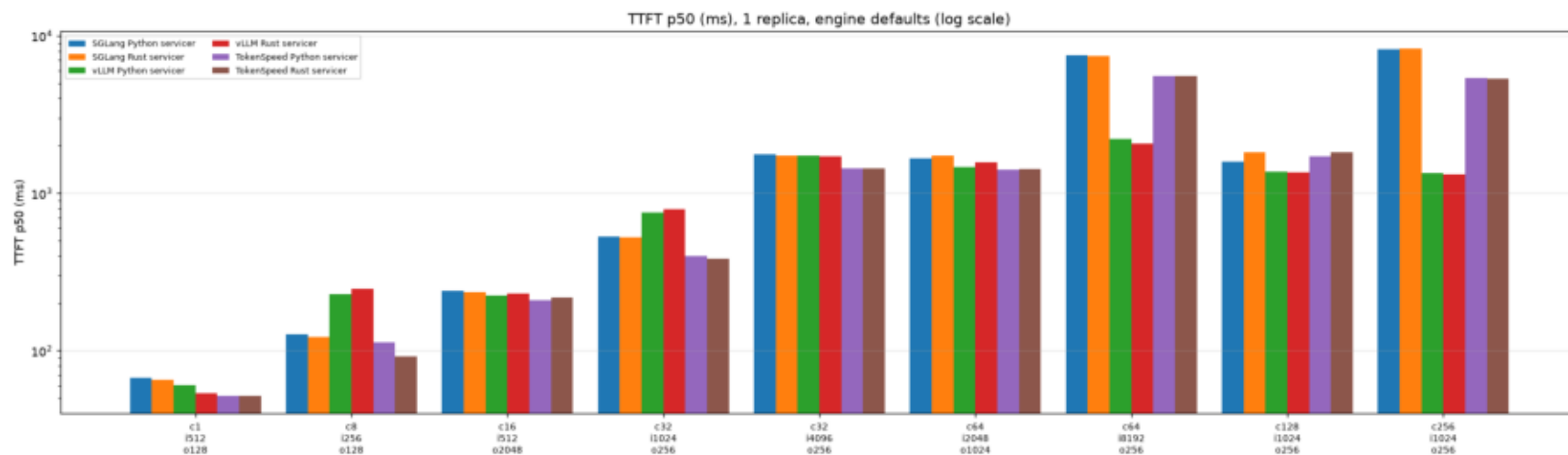
Output tokens / s, gRPC servicers behind smg gRPC mode, 1 replica



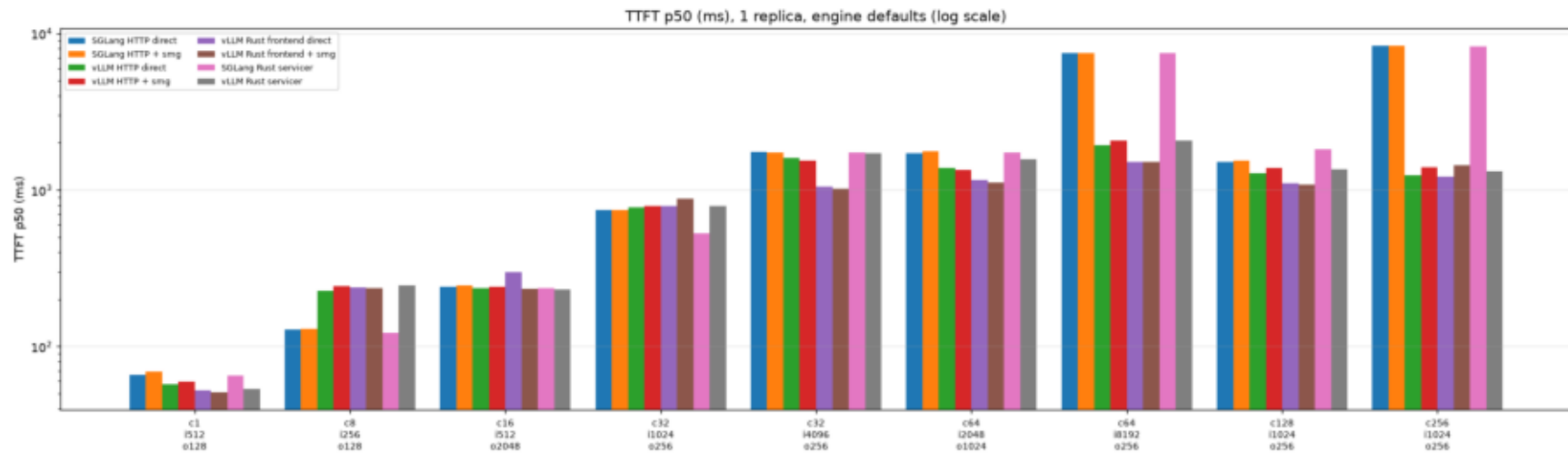
Output tokens / s, native servers direct vs behind smg vs smg gRPC with the Rust servicer, 1 replica



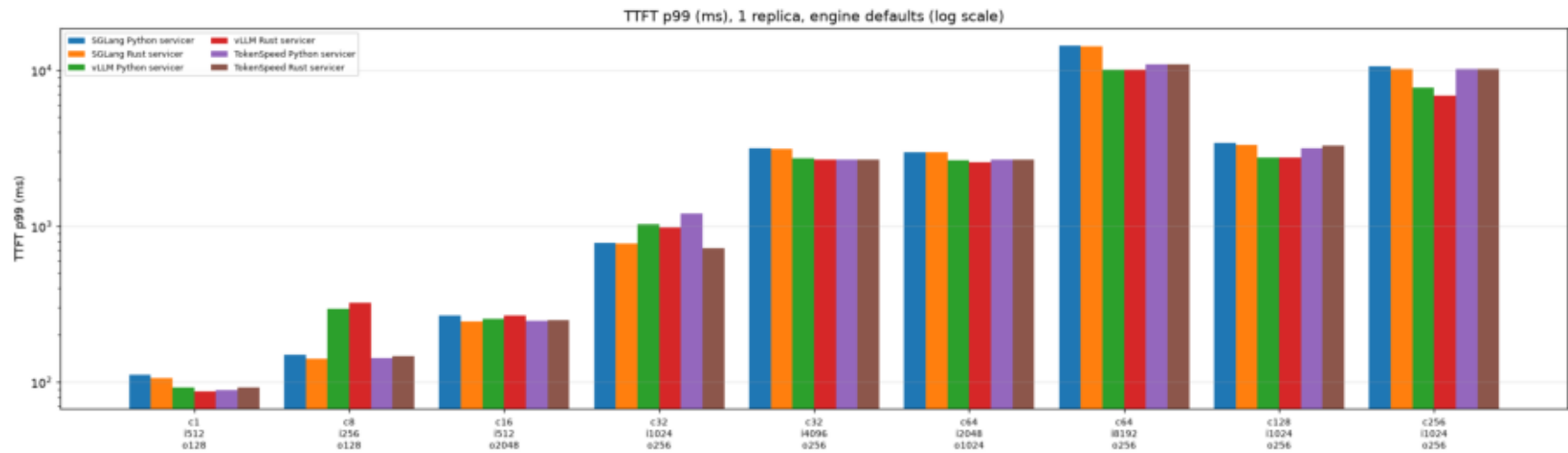
TTFT p50 (ms), gRPC servicers behind smg gRPC mode, 1 replica



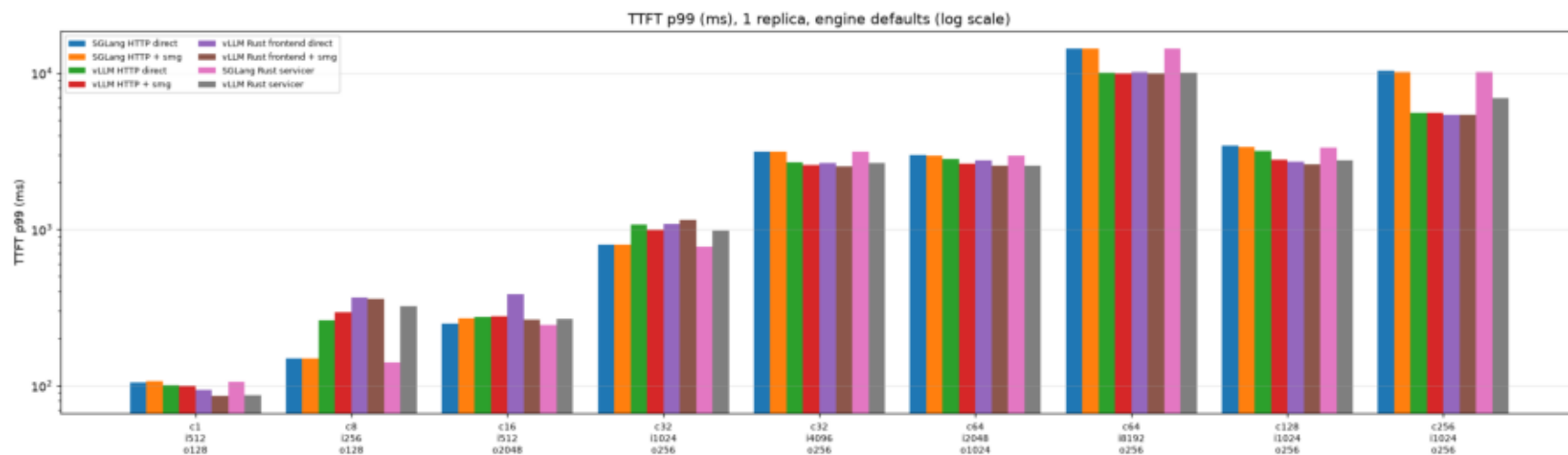
TTFT p50 (ms), native servers direct vs behind smg vs smg gRPC with the Rust servicer, 1 replica



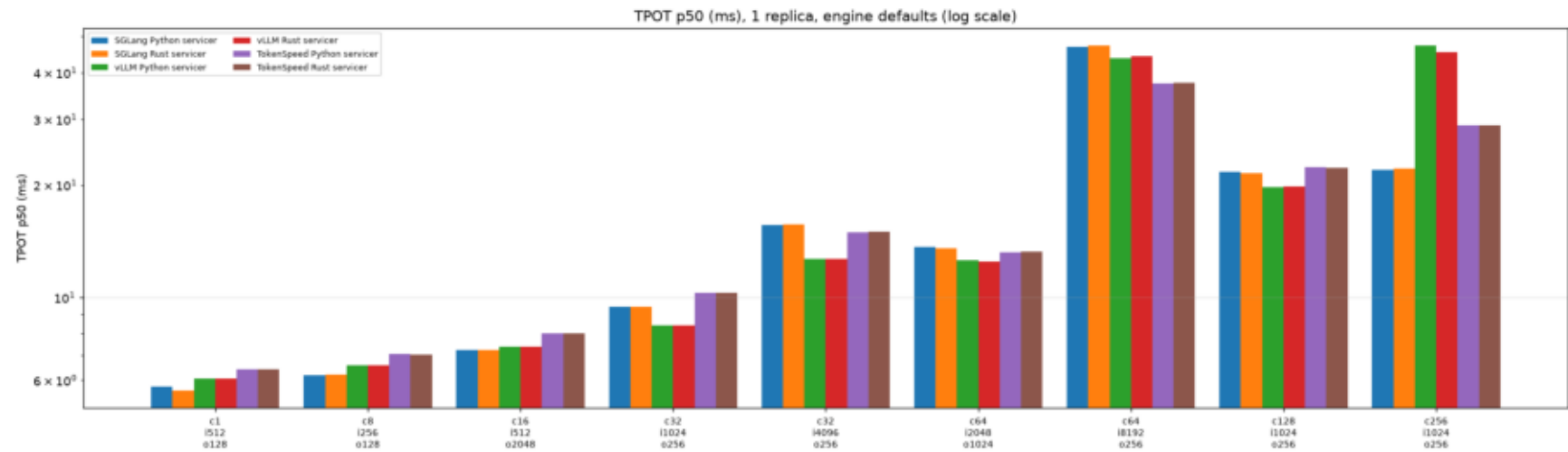
TTFT p99 (ms), gRPC servicers behind smg gRPC mode, 1 replica



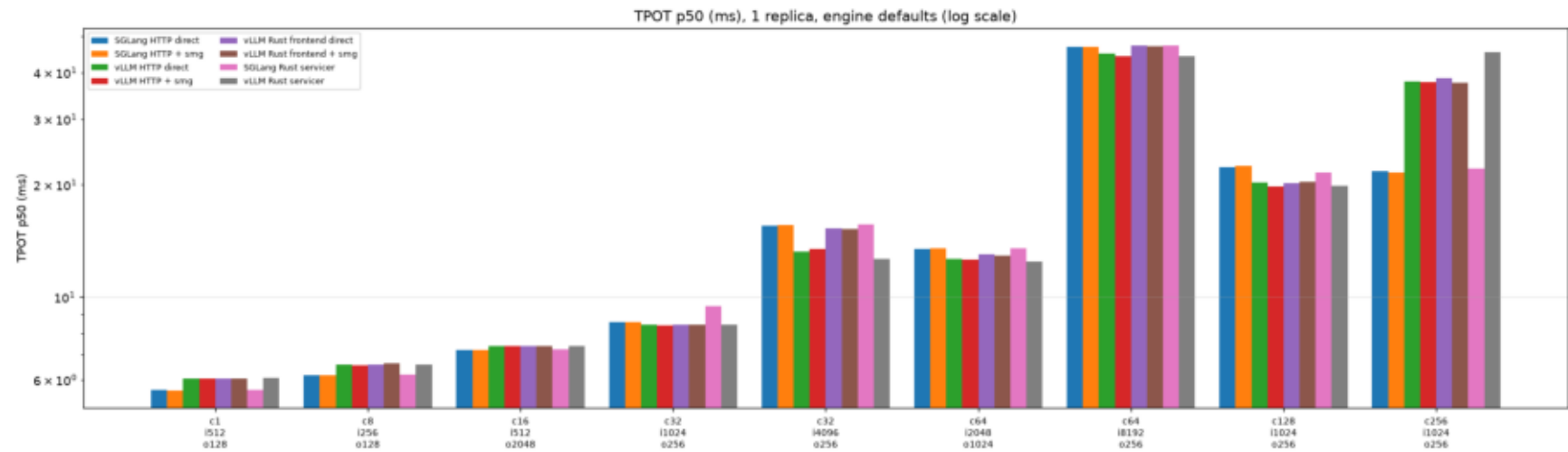
TTFT p99 (ms), native servers direct vs behind smg vs smg gRPC with the Rust servicer, 1 replica



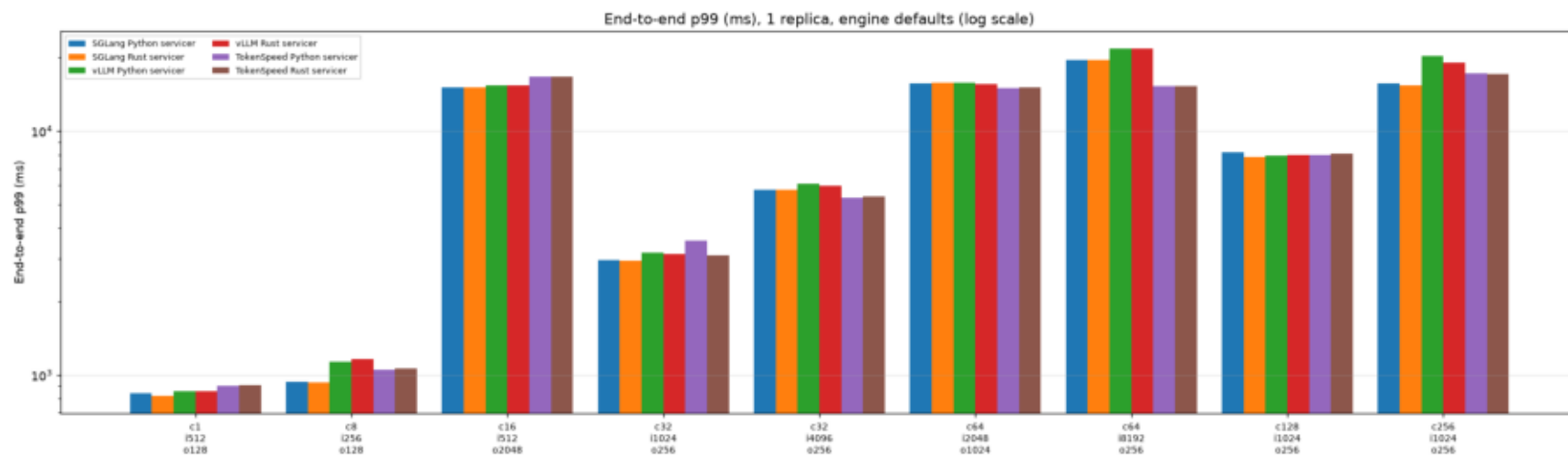
TPOT p50 (ms), gRPC servicers behind smg gRPC mode, 1 replica



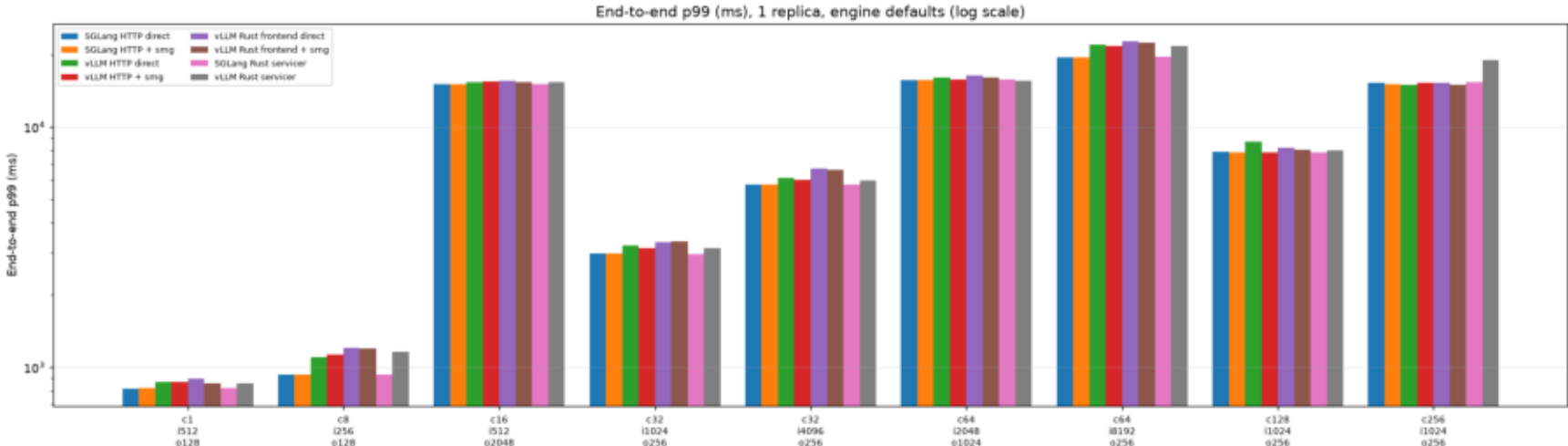
TPOT p50 (ms), native servers direct vs behind smg vs smg gRPC with the Rust servicer, 1 replica



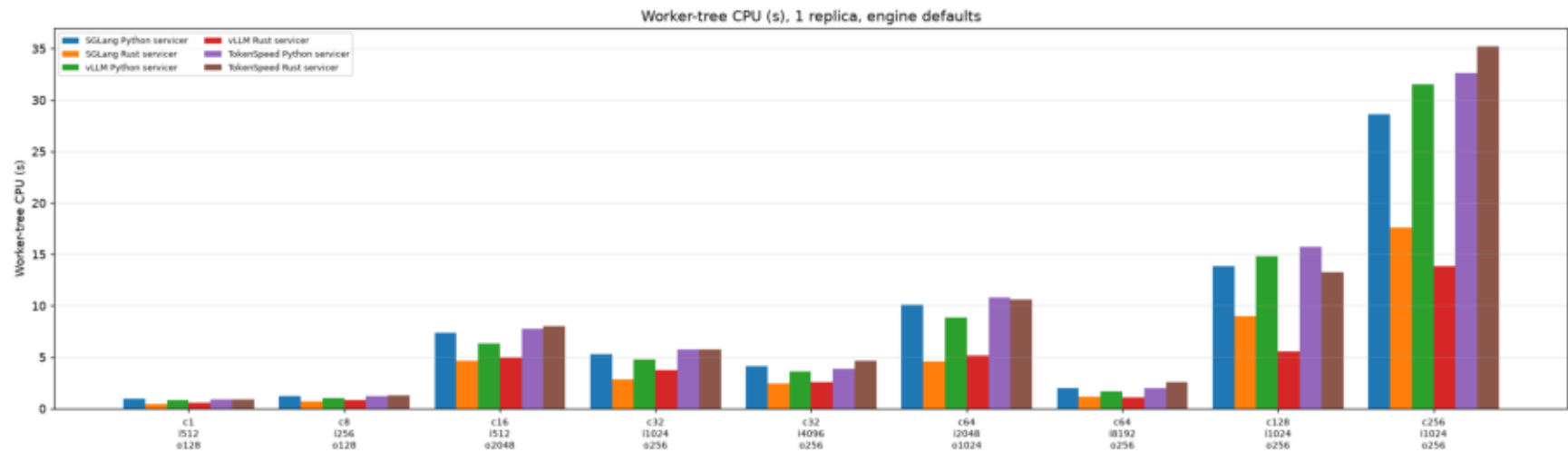
End-to-end p99 (ms), gRPC servicers behind smg gRPC mode, 1 replica



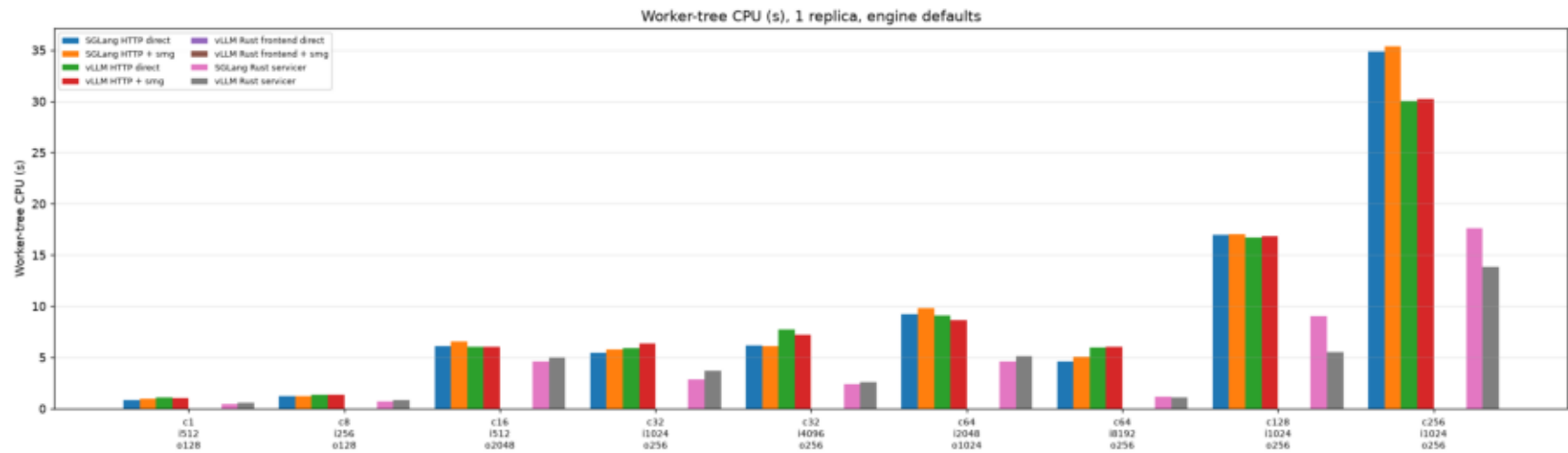
End-to-end p99 (ms), native servers direct vs behind smg vs smg gRPC with the Rust servicer, 1 replica



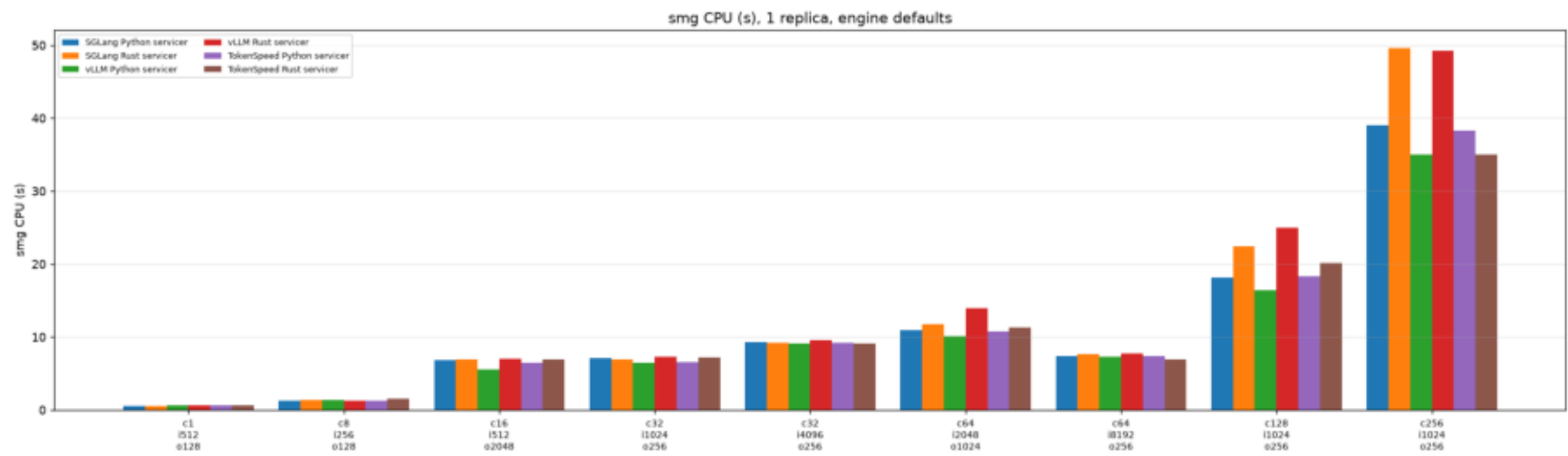
Worker-tree CPU (s), gRPC servicers behind smg gRPC mode, 1 replica



Worker-tree CPU (s), native servers direct vs behind smg vs smg gRPC with the Rust servicer, 1 replica



smg CPU (s), gRPC servicers behind smg gRPC mode, 1 replica



smg CPU (s), native servers direct vs behind smg vs smg gRPC with the Rust servicer, 1 replica

